Juan Pablo Roldan

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PROFILE

PhD student in aerospace engineering specializing in hypersonic and multiphase flows. Develops numerical methods for shock-driven droplet breakup and high-speed multiphase transport, with application to hypersonic rain environments, entry aerothermodynamics, and liquid-fueled propulsion. Complementary focus on GPU parallelization, reduced-order and data-driven modeling, and scientific software development for large-scale simulation. Seeking to pair computational work with experimental validation in high-speed flow facilities.

EDUCATION

PhD Aerospace Engineering, Aerodynamics and Propulsion — *Aug 2026 – IP*
Embry-Riddle Aeronautical University (ERAU), Daytona Beach, FL, USA

MS Aerospace Engineering, Aerodynamics and Propulsion (Dual Degree, GPA 4.00) — *Aug 2024 – Aug 2026*
Embry-Riddle Aeronautical University (ERAU), Daytona Beach, FL, USA
Universidad del Valle (UV), Cali, Colombia

BS Aerospace Engineering, Minor in Computer Science (GPA 4.00) — *Aug 2021 – May 2024*
University of Central Florida (UCF), Orlando, FL, USA

CONFERENCE PAPERS

Juan P. Roldan, Brendon A. Cavainolo, Sheryl M. Grace, Michael P. Kinzel, Guillermo A. Jaramillo, "Initial Perturbation Effects on Surface Instability Wavelength Selection in High-Mach Drop Aerobreakup," *AIAA SciTech, Jan 2027. (Abstract submitted)*

Juan P. Roldan, Sutharsan Satcunanathan, Georgios Goinis, "POD Airfoil Dimensionality Reduction for Compressor Cascade Optimization," *AIAA SciTech, Jan 2027. (Abstract submitted)*

Juan P. Roldan, Brendon A. Cavainolo, Rebecca C. Shannon, Sheryl M. Grace, Michael P. Kinzel, "Effect of Entrapped Non-Condensable Gas on Shock-Driven Droplet Deformation: Implications for Hypersonic Rain Environments," *AIAA International Space Planes and Hypersonic Systems and Technologies Conference, Jul 2026.*

Eric M. Yoerg, **Juan P. Roldan**, Donald Tuten, Michael Potash, Rui Harrison, Kassandra J. Durst, Michael P. Kinzel, Surabhi Singh, "Development of an Exploding Wire Facility to Study Shock Propagation at Embry-Riddle Aeronautical University," *AIAA SciTech, Jan 2025.*

RESEARCH EXPERIENCE

Hypersonic and High-Speed Flows

Numerical Study of Shock-Induced Droplet Deformation with Cavitation Nuclei Seeding — *Aug 2024 – IP*
Graduate Researcher, ERAU Computational Fluids and Aerodynamics Laboratory (M. P. Kinzel), and Universidad del Valle, IMPETUS INDOMITUS (G. A. Jaramillo)

- Simulating shock-driven droplet deformation and aerobreakup using the Multi-Component Flow Code (MFC), resolving bubble dynamics from seeded cavitation nuclei and entrapped non-condensable gas.
- Quantifying how initial perturbations govern surface instability wavelength selection during high-Mach aerobreakup, with implications for hypersonic rain erosion environments.
- Developing linear stability analysis capability and evaluating paths to incorporate data-driven and reduced-order methods.

Enhanced Backshell Heating Estimates for Dragonfly's Aeroshell — *Jun 2023 – Aug 2023*
Visiting Researcher, NASA Ames Research Center, Aerothermodynamics Branch | NASA MUREP-STAR

- Simulated chemically reacting hypersonic flow over the Dragonfly aeroshell during Titan entry using US3D, targeting the separated backshell wake where heating estimates are most grid-sensitive.
- Built and evaluated a metric-based mesh adaptation workflow to improve backshell heating predictions without uniform grid refinement.
- Completed as a 10-week embedded placement at a national research center, working within an operational aerothermodynamics team.

Hypersonic Waverider Design and Analysis Tool — *Jan 2025 – May 2025*
Lead Developer | github.com/JuanPabloRoldan/HyPyRider

- Led development of a Python-based rapid design and analysis tool for waverider geometries, following the design methodology of K. G. Bowcutt's thesis.

Multiscale Simulations of Scramjet Inlets with Dispersed Particles *May 2024 – Aug 2024*

Undergraduate Researcher, UCF Department of Mechanical and Aerospace Engineering (M. P. Kinzel)

- Conducted 2D steady non-reacting simulations of scramjet inlets under nominal flow conditions.
- Captured droplet trajectories and local flow variables via a dispersed multiphase model, supplying initial and boundary conditions for microscale single-droplet simulations.

Reduced-Order Modeling and Spacecraft Applications

POD Dimensionality Reduction for Compressor Cascade Optimization *Jun 2025 – Aug 2025*

Visiting Researcher, DLR Cologne, Institute of Propulsion Technology, Fan and Compressor Group | NSF IRES

- Investigated proper orthogonal decomposition for reducing design-space dimensionality in compressor cascade optimization.
- Characterized the tradeoff between dimensionality reduction and loss of optimization accuracy.
- Completed as a 10-week embedded placement at a national aerospace research center abroad.

Plume Impingement Analysis: RPOD Maneuvers to the Lunar Gateway *Sep 2023 – May 2024*

Co-Lead Developer | github.com/plume-kit/PyRPOD | Mentor: J. S. Pitt

- Co-led development of a rapid-analysis tool characterizing visiting-vehicle thruster configurations by impingement, fuel usage, and safety metrics.
- Implemented rarefied gas dynamics and gas-surface interaction models.

INDUSTRY-PARTNERED RESEARCH DEVELOPMENT

Modernization of CFD Tools for Advanced Propulsion Applications (Not Awarded) *Submitted May 2026*

NASA STTR Phase I | Partners: Advanced CFD Solutions, Worcester Polytechnic Institute

- Led technical development of the proposal concept, centered on multiphase-flow modeling for liquid-fueled detonation and shock-droplet interaction applications.

Augmenting Classical HPC with Quantum Computing Capabilities for CFD (Not Awarded) *Submitted Sep 2025*

DoD DTRA STTR Phase I | Partners: Advanced CFD Solutions, ERAU; with support from Oak Ridge National Laboratory

- Led technical development of the proposal concept, targeting faster computational air-blast prediction by offloading portions of the CFD workload from classical HPC to quantum hardware.

GRANTS AND AWARDS

NSF International Research Experience for Students — $13,000 *Jun 2025 – Aug 2025*

Funded 10 weeks as a visiting researcher at DLR Cologne.

Benjamin A. Gilman International Scholarship — $6,000 *Jan 2024*

Funded a short-term study abroad program in Sao Paulo, Brazil.

NASA MUREP Scholarship for Training, Advancement, and Research (MUREP-STAR) — $10,000 *Mar 2023 – Aug 2023*

Funded a three-month placement as a visiting researcher at NASA Ames Research Center.

UCF National Hispanic Recognition Program Scholarship — $48,000 *Aug 2021 – May 2024*

Funded four years of undergraduate study in aerospace engineering.

SKILLS AND SOFTWARE

CFD Software: Multi-Component Flow Code (MFC), ALPACA (Technical University of Munich), US3D, STAR-CCM+

Programming: Python, C, MATLAB, Java, CUDA

HPC and Tools: Linux, SLURM, PBS, Git/GitHub

ADDITIONAL EXPERIENCE

Academic Exchange, Universidad del Valle, Cali, Colombia *Aug 2025 – May 2026*

- Completed coursework and research under the ERAU-UV dual-degree agreement; supported peer fellowship applications and English-language practice, and helped establish collaborative research networks.

Contemporary Challenges in Global Aerospace, UCF and Embraer, Sao Paulo, Brazil *Jan 2024 – May 2024*

Faculty-embedded short-term program on aerospace manufacturing design methods, with industry site visits.

High-Cycle Fatigue Testing, UCF Mechanical and Aerospace Engineering (J. L. Kauffman) *Sep 2022 – Dec 2022*

Designed a low-cost rig for fatigue testing of repaired turbomachinery blades; implemented a LabVIEW PID controller to hold target stresses.